Weekly Report

Agenda

- Data Capture
- Data Cleaning, Techniques
- Moving Forward

Data Capture

- 1 Data Capture at Lake Murray Spot
- Drone Pictures and Footage

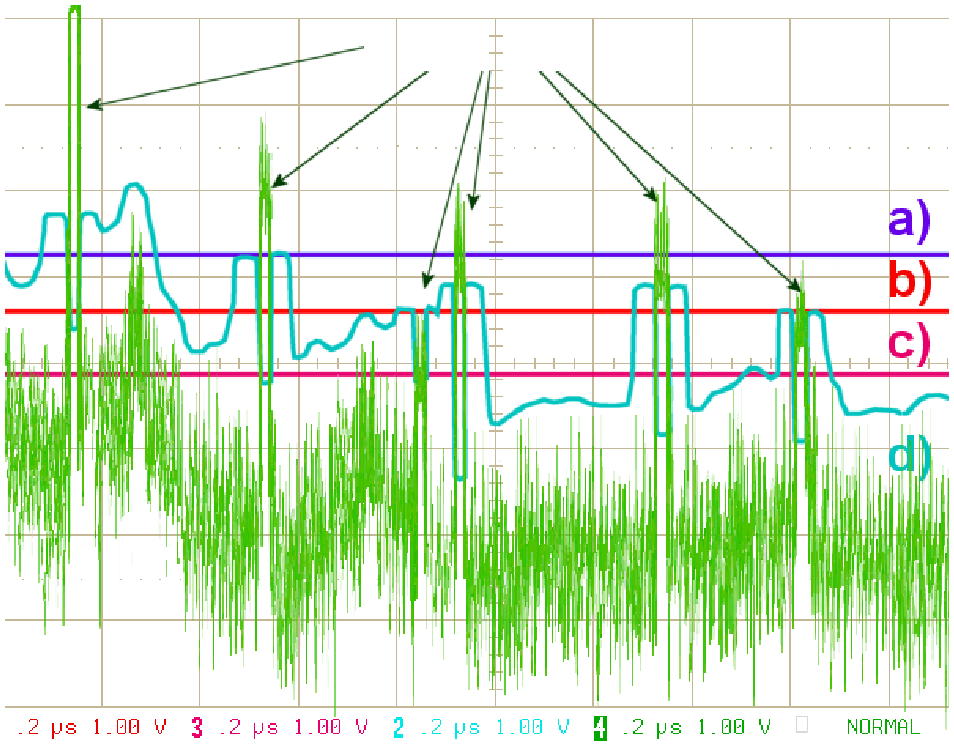
- Good Data?
- Good Pictures?
- Next Spot?

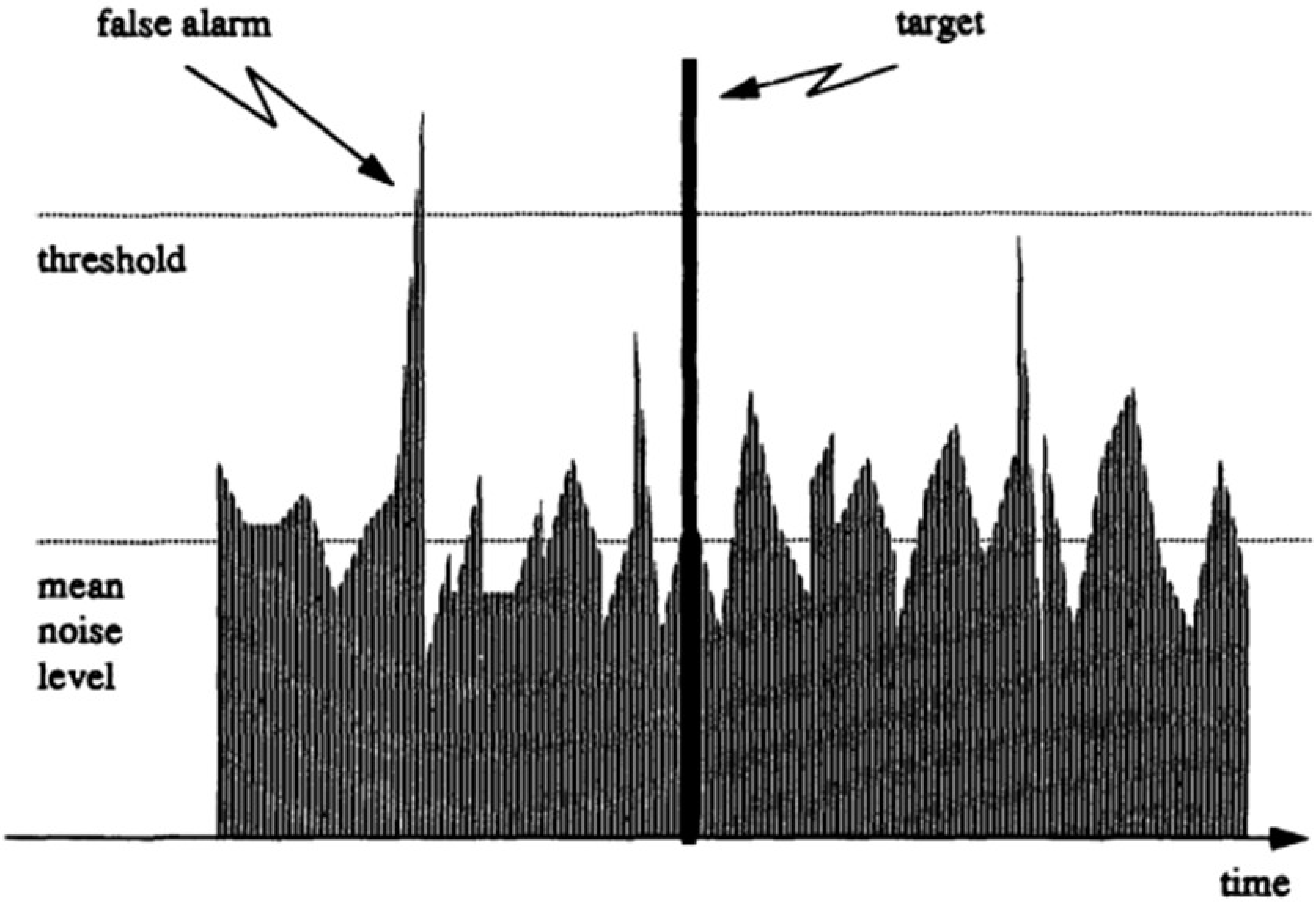
Constant False Alarm Rate Algorithms

CFAR Algorithms looks at surrounding radar returns to estimate noise level.

Instead of using a fixed threshold for detection, a CFAR Algorithms adjust the threshold (higher in noisier regions, lower in quieter) to catch the best images

The CFAR is a rule based algorithm that could give real-time detection of contacts on the radar.



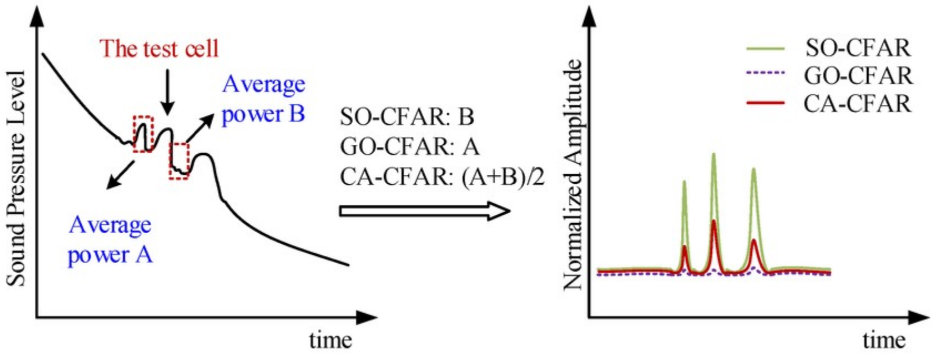

false alarm
target
threshold
mean
noise
level
time

GO-CFAR

Compares two reference windows before and after the Cell Under Test

Uses the higher of the two averages of the noise to set the threshold

Supposedly is the best for clutter edges for maritime use

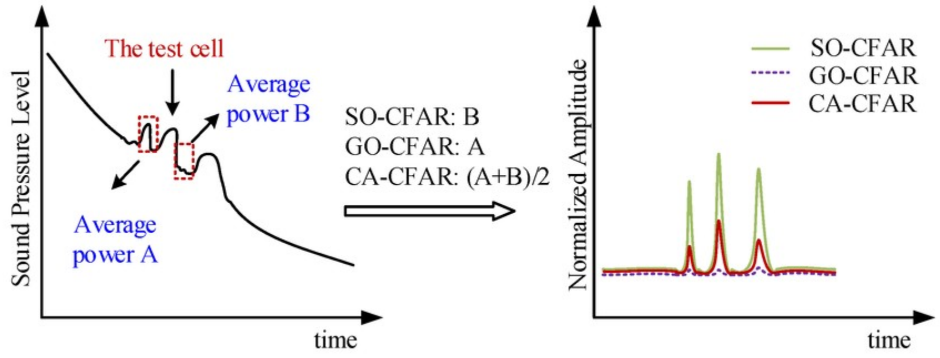


SO-CFAR

Compares two reference windows before and after the Cell Under Test

Uses the lower of the two averages of the noise to set the threshold

Supposedly is the best for multi target environments, like smaller targets near larger ones



Moving Forward

- Continue with weekly data capture at new locations
- Convert .csv files to .png radar images
- Test CFAR algorithms until an acceptable level of “Clean Data” is reached
- Use simple data augmentation to make more radar data to train AI model